

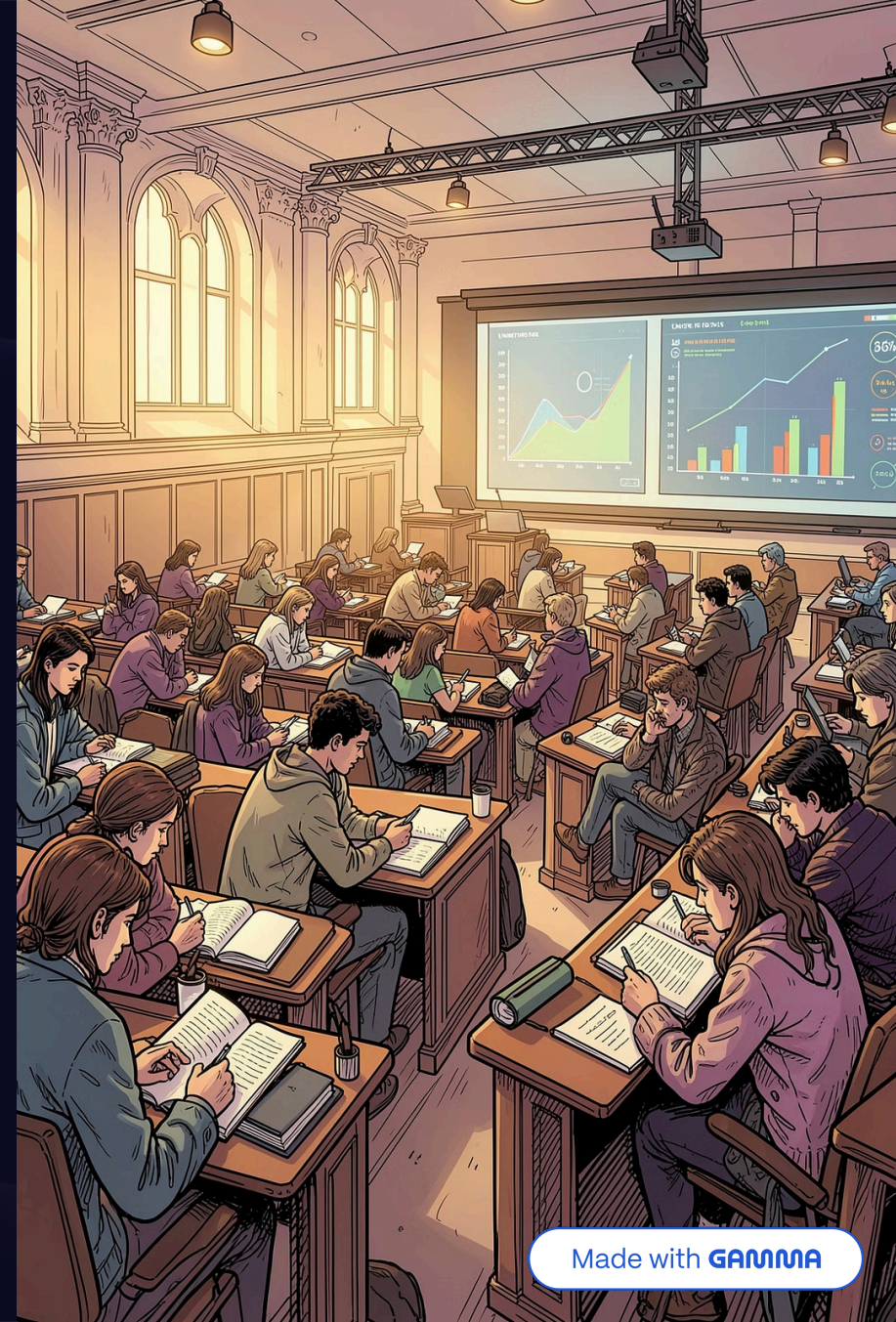
Learning Behavior Clustering and Student Profiling

An Unsupervised Machine Learning System for Early Identification
of At-Risk Students

MACHINE LEARNING · SPRING 2025–2026

ENSIAS

KAGGLE DATASET · 14,003 RECORDS



Made with GAMMA

The Challenge in Higher Education

Traditional grading masks behavioral patterns that could predict academic struggle **months in advance**.

Silent Failure

Students fail without early warning signals.

One-Size-Fits-All

Teaching ignores diverse learning profiles.

Blind Spots

Standard monitoring misses **30%** of at-risk students.

Disengaged Talent

High performers go under-challenged, risking disengagement.

i The Opportunity: Multi-dimensional behavioral analysis can identify student profiles and enable targeted interventions before failure occurs.



Research Question and Goals

Can unsupervised clustering on engineered behavioral features reliably identify meaningful student profiles actionable for academic intervention?

Core Objectives

- 01
- Preprocess **14,003** student records
- 02
- Engineer **13** behavioral features
- 03
- Compare **7** clustering algorithms
- 04
- Identify **2–4** distinct student profiles
- 05
- Build an early-warning system
- 06
- Provide data-driven recommendations

Success Criteria

- Silhouette Score > **0.08**
- Cluster profiles align with **FinalGrade**
- At-risk cluster detectable mid-semester
- System interpretable for educators

Data Source and Scope

KAGGLE · STUDENT PERFORMANCE & LEARNING STYLE



Dataset at a Glance

- **14,003** anonymized student records
- **16** raw features across academics, behavior, psychology, and demographics
- Academic: StudyHours (5–44 h/week), Attendance (60–100%), ExamScore (40–100)
- Psychological: Motivation, StressLevel, Resources (Low/Med/High)
- Outcome: FinalGrade (A/B/C/Fail)

Data Quality

- No missing values — 100% complete
- **1,534** duplicates removed (10.95%)
- **3** consistency rules → **225** flagged rows (retained)
- Final working dataset: **12,469 clean records**

The Machine Learning Pipeline

Feature Engineering

13 composite features; 24-dim matrix; standardize.

Model Selection

Tested 7 clustering algorithms; 3 metrics.

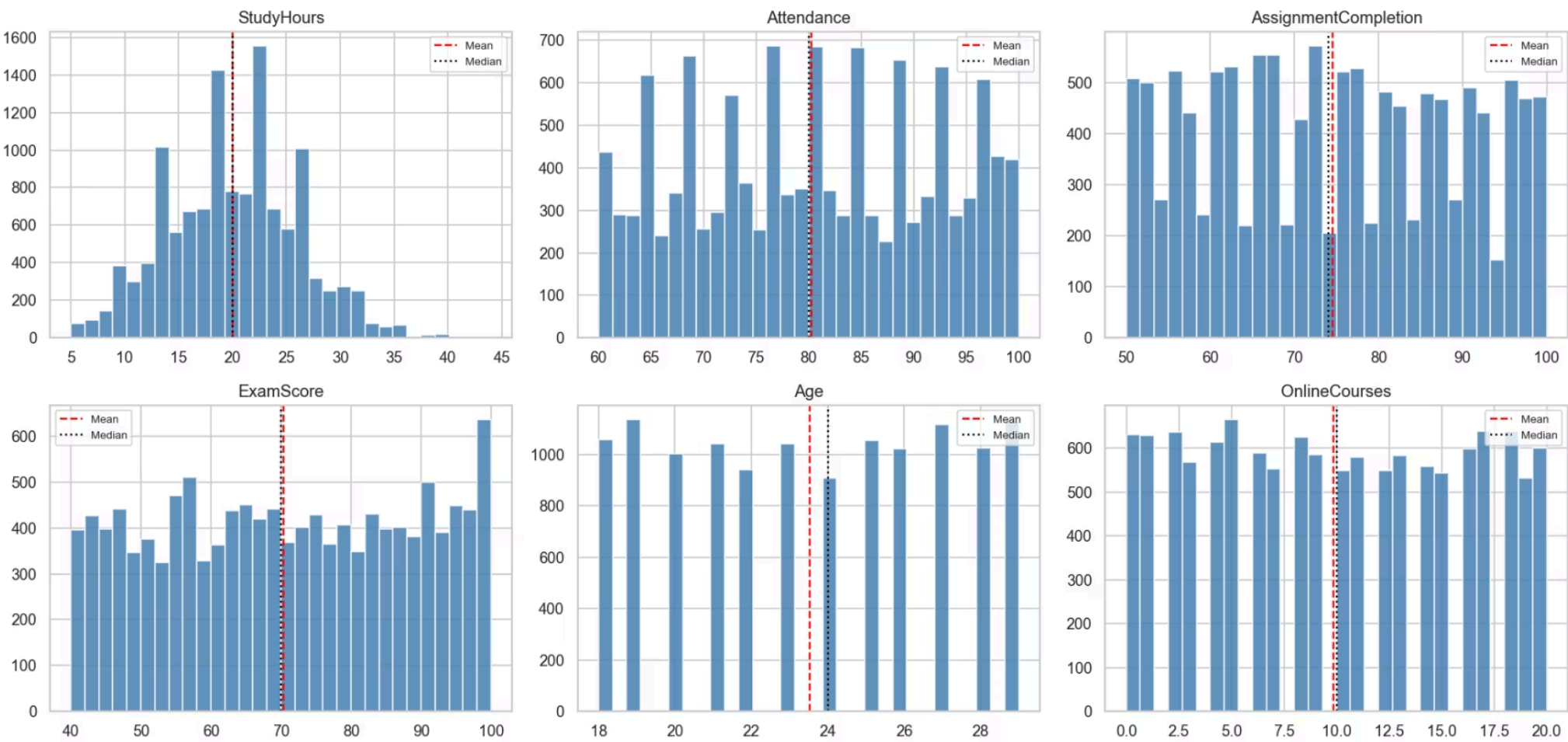
Data Preparation

Duplicate removal, EDA, outlier detection.

Profile Discovery

GMM chosen; identified 3 student clusters.

Distributions of Continuous Features



1

16 Raw Features

2

13 Engineered Features

3

24-Dimensional Matrix

4

Standardized & Ready

No single raw variable cleanly separated students (max correlation $r = 0.03$), making feature engineering and algorithmic clustering essential to uncovering meaningful behavioral patterns.

Why Raw Features Aren't Enough

A student studying **30 hours** but scoring poorly is fundamentally different from one studying **10 hours** and scoring the same. Raw variables hide this distinction.



StudyEfficiency

ExamScore ÷ **StudyHours** — identifies high-return learners vs. inefficient effort



EngagementScore

Weighted blend of **Attendance**, **Assignments**, **Discussions**, **Extracurricular**



WellbeingIndex

Resources + **Motivation** – **StressLevel** — balances support, drive, and burden



PerformanceGap

ExamScore – **AssignmentCompletion** — separates test-savvy students from consistent performers



Key insight: Raw attendance looks normal for struggling students — but **StudyEfficiency** + **PerformanceGap** reveals ineffective learning that conventional systems miss entirely.



Why Gaussian Mixture Models Win

Algorithm	Silhouette	DBI	Verdict
GMM — Selected	0.111	2.326	Best performer
K-Means	0.078	2.819	Assumes spherical clusters
HDBSCAN	0.096	4.290	Only 2 clusters, 9% noise
BIRCH	0.066	3.413	Lower cohesion
Hierarchical	0.058	3.495	Subjective cut-level
DBSCAN	NaN	NaN	Too uniform density
Mean Shift	NaN	NaN	Collapsed to 1 cluster

Why GMM is Superior

- **Soft Assignments:** Probability per cluster — honest about borderline cases
- **No Geometric Assumptions:** Captures natural behavioral overlaps
- **Probabilistic Framework:** Enables confidence-based warning thresholds
- **Interpretability:** 3 components map directly to educational categories

What the Data Revealed



Cluster 0 — Consistent Average Performers

85.0% · 10,599 students

Work hard but use inefficient strategies. Invisible to support services — yet structured study skills coaching could unlock significant gains.



Cluster 1 — Last-Minute High Performers

7.2% · 899 students

Exceptional efficiency on minimal hours. Under-challenged by current curriculum — enrichment and peer mentoring would develop their potential.



Cluster 2 — Struggling / At-Risk

7.8% · 971 students

Attend class and submit work but learn ineffectively. **Conventional systems would not flag them.** This is the system's most critical finding.

From Data to Decision-Making

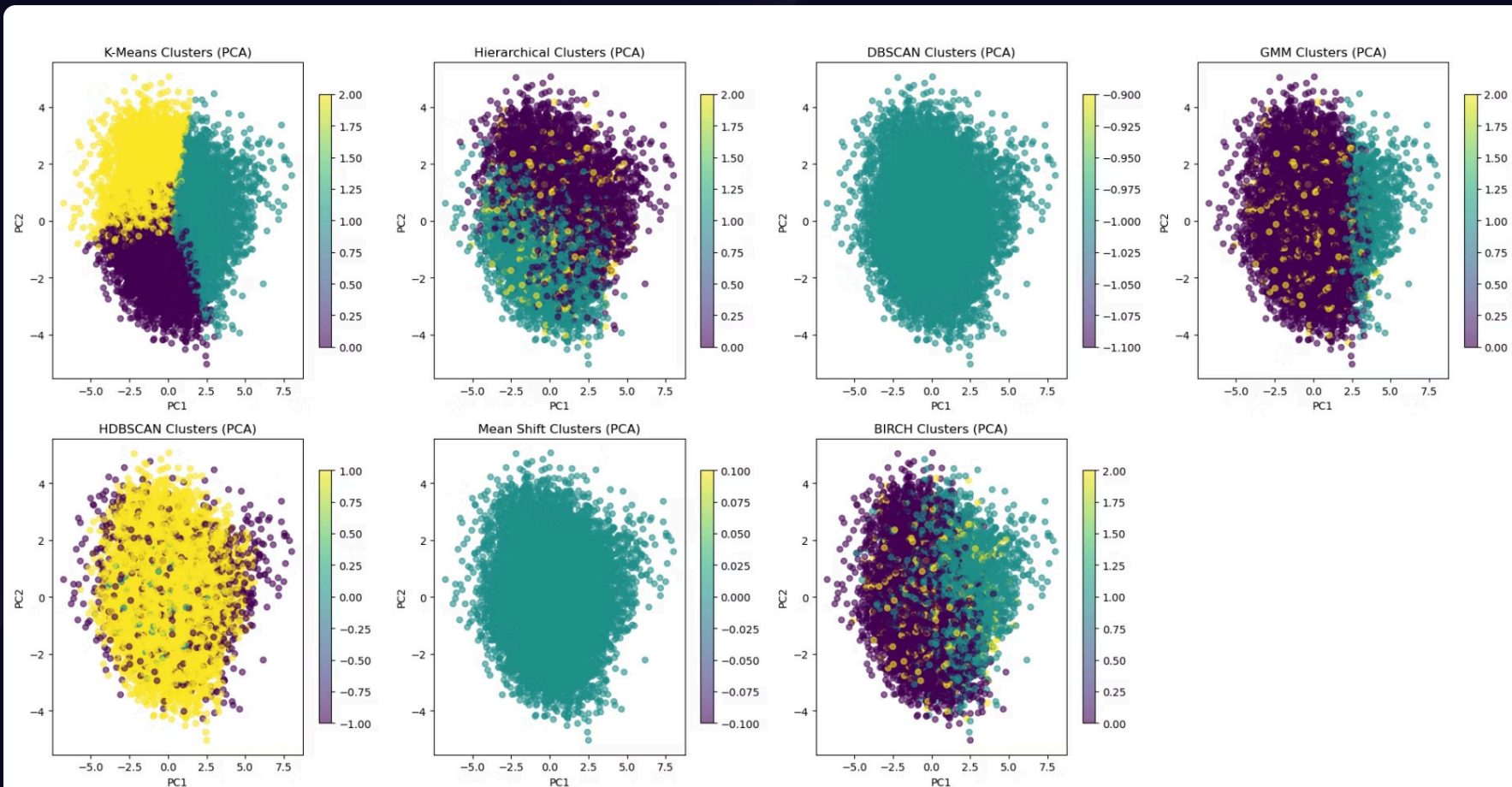


Figure 2: PCA projections of cluster assignments across all seven algorithms tested

System Deployment Principles

Integrate

Embed in academic
advising dashboards

Human-in-the- Loop

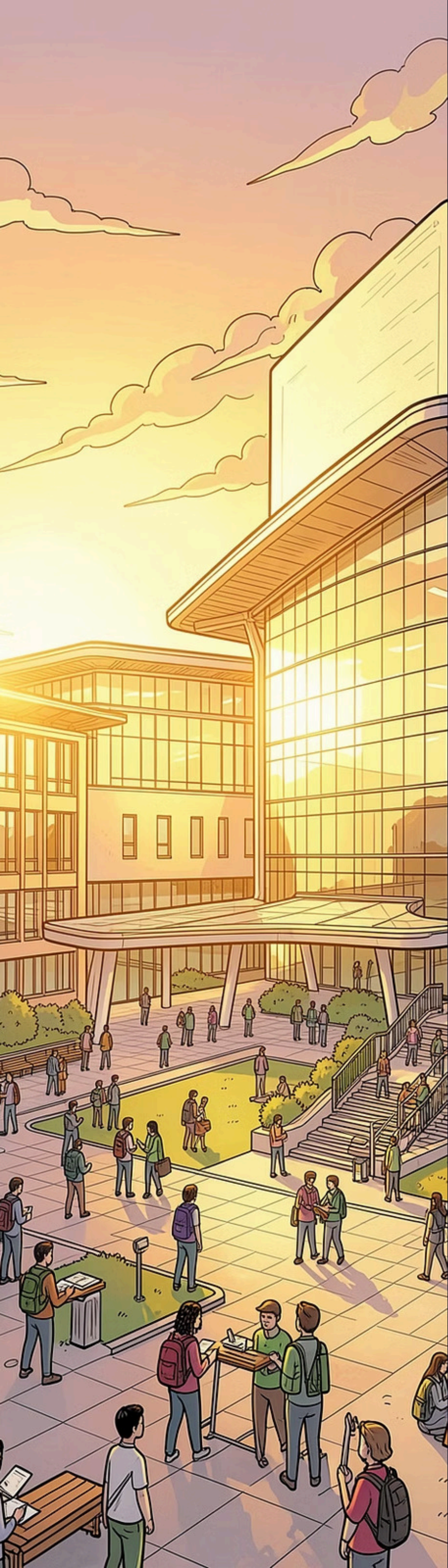
Advisor judgment +
algorithmic flags

Mid-Semester

Monitor before grades
collapse

Iterate

Track intervention
outcomes over time



Summary and Impact

14,003

Records Analyzed

Zero missing values

13

Features Engineered

Revealing hidden patterns

7

Algorithms Tested

GMM selected as optimal

971

At-Risk Students Found

Invisible to conventional metrics

Efficiency Over Effort

Study efficiency matters more
than raw hours

Silent Majority Improvable

85% of students can benefit
from targeted coaching

Proactive, Not Reactive

Intervene before students struggle

Future Directions: Temporal tracking across semesters, integration with institutional systems, personalized interventions by cluster, and validation across additional institutions.